

Week 8 – Workshop

Geothermics: Steady-State Heat Flow



Geophysics 3A: Potential Fields & Geothermics

July 6, 2026

Duration: 3 hours

Setting: Workshop room

Preparation

Pre-reading:

Lecture videos:

Notes:

Purpose

This workshop uses conservation-of-energy reasoning to build intuition for steady-state geotherms: how thermal conductivity contrasts refract isotherms and redistribute gradients, and how crustal heat production controls geotherm curvature and deep temperatures.

Learning Outcomes

By the end of this workshop, students should be able to:

- Use conservation of energy to reason about steady-state geotherms.
- Explain how thermal conductivity contrasts redistribute isotherms.
- Interpret geotherm curvature in terms of crustal heat production.
- Relate conductivity and heat production variations to geothermal potential.

Session Flow (180 min)

0–15 min: Conceptual Framing

- Steady-state heat flow as an energy bookkeeping problem.
- Review Fourier's law and the meaning of heat flow.
- Emphasize the separation of roles:
 - conductivity controls *gradients*,
 - heat production controls *curvature*.

15–45 min: Activity A: Refraction Exercise I: Insulating Sediments

Setup Compare two crustal columns with identical basal heat flow:

- Case A: crystalline crust exposed at the surface.
- Case B: crystalline crust overlain by a low-conductivity sedimentary layer.

Activities

- Predict temperature gradients in sediments vs basement.
- Sketch qualitative isotherms for each case.
- Identify where isotherms compress and where they stretch.

Key Concepts

- Heat flow conserved across layers.
- Lower conductivity $\rightarrow$ steeper gradient.
- Sediments act as a thermal blanket.

Geological Discussion

- Implications for geothermal gradients measured in boreholes.
- Relative geothermal potential of sedimentary basins vs exposed cratons.

45–80 min: Activity B: Refraction Exercise II: Layered Crustal Materials

Focus Thermal refraction arising from conductivity contrasts within the crust.

Examples

- Shales (low k) vs quartzites (high k).
- Cyclic sedimentary layering preserved through metamorphism.

Activity

- Apply $q = -k\nabla T$ across layered media.
- Examine how gradients vary layer by layer.
- Discuss effective anisotropy in cyclically layered strata.

Connection

- Tie back to Week 2 physical-properties mixing demos.
- Discuss refraction in tilted and folded strata.

80–90 min: Break

90–125 min: Activity C: Energy Conservation and Geotherm Curvature

Central Question How can two terranes with identical surface heat flow have different Moho temperatures?

Steps

- Write the energy balance:

$$q_0 = q_b + \int_0^{z_c} A(z) dz$$

- Infer relative crustal heat production given different basal inputs.
- Introduce the steady-state solution and the curvature term:

$$T(z) = T_0 + \frac{q_0}{k}z - \frac{A}{2k}z^2$$

Key Insight Curvature differences imply different deep temperatures even when surface gradients match.

125–160 min: Activity D: Why Is Archean Heat Production So Low?

Observation Decay alone predicts high early heat production, yet preserved Archean crust shows limits.

Guided Discussion Topics

- Differentiation and partitioning of LILEs (K, Th, U).
- Selective erosion of felsic upper crust.
- Reworking: remelting and redistribution of older crust.
- Foundering of dense mafic roots.
- Secular mantle thermal evolution.

160–180 min: Synthesis

- Summarize roles of k and A .
- Connect observations to upcoming steady-state problem set.
- Preview transition to transient heat flow.

Activity A

Thermal Refraction by Insulating Sediments

Purpose

To build intuition for how low-conductivity surface layers redistribute temperature gradients without changing the total heat flow.

Geological Context

Consider two continental regions with identical basal heat flow from the mantle:

- Region A: crystalline crust exposed at the surface.
- Region B: crystalline crust overlain by a thick sedimentary basin.

The sediments are dominated by shales and porous clastic rocks with lower thermal conductivity than basement rock.

Assumptions

- Steady-state heat flow.
- No internal heat production in sediments.
- Heat flow is vertical and laterally uniform.

Tasks

1. Using Fourier's law,

$$q = -k \frac{dT}{dz},$$

explain qualitatively how the temperature gradient differs between sediments and basement.

2. Sketch temperature versus depth for both regions on the same axes. Clearly label:
 - sediment thickness,
 - basement,
 - relative steepness of gradients.
3. Indicate where isotherms are compressed and where they are stretched.
4. Which region would exhibit a higher temperature at the sediment–basement interface? Explain why.

Discussion

- How does the presence of sediments affect measured geothermal gradients?
- Which region would you expect to have higher geothermal potential?
- Why might sedimentary basins be favorable geothermal targets?

Activity B

Thermal Refraction in Layered Crustal Rocks

Purpose

To explore thermal refraction caused by conductivity contrasts within the crust, particularly in sedimentary and metamorphic layering.

Geological Context

Many crustal sections contain cyclic layering inherited from sedimentary environments, such as alternating shales and quartz-rich sandstones that later become quartzites.

- Shales: low thermal conductivity.
- Quartzites: high thermal conductivity.

This layering can persist through burial, deformation, and metamorphism.

Tasks

1. Consider a vertical sequence of alternating shale and quartzite layers. Assume constant heat flow through the sequence.
2. Using

$$q = -k\nabla T,$$

explain how the temperature gradient differs in shale layers compared to quartzite layers.

3. Sketch a qualitative temperature–depth profile showing changes in slope across layers.
4. Now consider the same layered sequence tilted relative to vertical heat flow. Describe how thermal refraction will distort isotherms.

Extension

- How can cyclic layering create an effectively anisotropic thermal conductivity?
- How might folding or tilting further modify the thermal field?

Connection

Relate this behavior to the physical-properties mixing demonstrations discussed earlier in the course.

Activity C

Energy Conservation and Geotherm Curvature

Purpose

To understand how crustal heat production alters lithospheric temperatures even when surface heat flow is unchanged.

Scenario

Two continental terranes have identical surface heat flow, q_0 , but different amounts of heat supplied from the mantle at the base of the crust.

Tasks

1. Write the steady-state energy balance for the crust:

$$q_0 = q_b + \int_0^{z_c} A(z) dz$$

2. If Terrane A has a larger basal heat flow (q_b) than Terrane B, which must have higher integrated crustal heat production? Explain.
3. Consider the steady-state geotherm:

$$T(z) = T_0 + \frac{q_0}{k} z - \frac{A}{2k} z^2$$

Identify which term controls curvature.

4. Without performing calculations, explain how greater curvature affects:
 - temperature at the Moho,
 - depth of intersection with the mantle adiabat.

Key Insight

Explain why surface gradients alone are insufficient to determine deep crustal temperatures.

Activity D

Archean Heat Production

Purpose

To connect radioactive decay, crustal differentiation, and geological processes to the observed heat production of ancient continental crust, we'll investigate the following question "Why is Archean heat production so low?"

Radiogenic decay

The concentration of a radioactive isotope is given by

$$C(t) = C_0 e^{-\lambda t},$$

and the corresponding heat production is

$$A(t) = \sum_i H_i \lambda_i C_i(t),$$

where H_i is the heat released per decay and λ_i is the decay constant.

Observation

Simple decay arguments predict higher heat production in the Archean than today, yet preserved Archean crust shows relatively modest values.

Tasks

For each mechanism below, explain qualitatively how it could reduce the heat production of preserved crustal rocks:

1. Differentiation and melt extraction of LILEs (K, Th, U).
2. Selective erosion of felsic upper crust.
3. Reworking: remelting and redistribution of existing crust.
4. Foundering of dense mafic lower crust or arc roots.
5. Secular evolution of mantle temperature.

Differentiation

Concept: Heat-producing elements (K, Th, U) behave as incompatible elements and strongly partition into melts.

Let $D \ll 1$ be the bulk partition coefficient of a heat-producing element. For fractional melting,

$$C_{\text{solid}} = C_0 (1 - F)^{(1/D-1)},$$

where F is the melt fraction. You could think of this simply as:

$$C_{\text{crust}} \approx \frac{C_{\text{mantle}}}{D} F$$

Selective erosion

Concept: Heat production is concentrated in felsic upper crust, which is preferentially eroded.

Treat the crust as two reservoirs. Let A_u be upper crust heat production. If erosion removes heat-producing material at rate E ,

$$\frac{dA_u}{dt} = -EA_u.$$

Solution:

$$A_u(t) = A_{u0}e^{-Et}$$

Reworking

Reworking here will be defined as reheating—past the closure temperatures of geochronometers—and remelting of existing crust rather than net addition of new crust.

Concept: Old crust is repeatedly melted, mixed, and redistributed during later tectonic events.

Schematic formulation: Model reworking as a mixing process. Let a fraction R of crust be reprocessed per event.

$$A_{n+1} = (1 - R)A_n + RA_{\text{new}},$$

where A_{new} reflects heat production typical of newly generated crust.

Foundering

Concept: Dense, mafic, heat-poor material is preferentially removed from the lithosphere.

Let the lower crustal mass is reduced over time. If mafic lower crust is removed at rate F_d ,

$$\frac{dM_{lc}}{dt} = -F_d M_{lc}.$$

The average crustal heat production becomes:

$$\bar{A}(t) = \frac{A_u M_u + A_{lc} M_{lc}(t)}{M_u + M_{lc}(t)}.$$

Secular cooling

Concept: The mantle was hotter early, affecting melt production rather than radiogenic inventory.

Mantle potential temperature declines approximately as:

$$T_m(t) = T_{m0}e^{-t/\tau},$$

where $\tau \sim 1\text{--}2$ Ga is a thermal timescale. Melt production gives:

$$F(t) \propto T_m(t) - T_{\text{solidus}}.$$

Synthesis Questions

1. Which processes change the *distribution* of heat-producing elements?
2. Which processes affect what crust is *preserved*?
3. Why does radioactive decay alone fail to predict observed Archean heat production?

Which processes primarily:

Process	Primary Physical Mechanism	Effect on Heat-Producing Elements (K, Th, U)	Primary Geological Consequence
Radiogenic decay	Exponential decay of radioactive isotopes	Monotonic decrease of heat production with time	Higher heat production in the past; explains first-order time trend only
Differentiation and melt extraction	Incompatible elements partition strongly into melts ($D \ll 1$)	Rapid transfer of LILEs from mantle to crust early in Earth history	Early enrichment of crust; depletion of mantle
Selective erosion	Preferential removal of felsic, heat-rich upper crust	Progressive loss of the most heat-producing crustal components	Older crust appears cooler than expected from decay alone
Reworking (remelting and redistribution)	Repeated partial melting and crustal mixing	Dilution and homogenization of extreme heat production signatures	Preserved Archean crust is rarely pristine
Foundering of mafic lower crust	Removal of dense, heat-poor lower lithosphere	Bias toward preservation of felsic, buoyant crust	Thermally stable, thick continental lithosphere
Secular mantle thermal evolution	Early high mantle temperatures enhance melt production	Controls efficiency of LILE extraction rather than inventory	Strong early differentiation followed by declining crustal growth

- redistribute heat-producing elements?
- remove heat-producing material from the crust?
- bias what crust survives to the present day?

Reflection

Why is radioactive decay alone insufficient to explain Archean heat production patterns?

Concept Takeaway: Heat Production, Curvature, and Preservation

Radiogenic decay alone predicts that heat production should decrease smoothly with time, but observed Archean crust shows much lower heat production than this simple expectation.

The heat production of preserved continental crust reflects the combined effects of:

- *redistribution* of heat-producing elements through differentiation and melting,
- *removal* of heat-rich material by erosion and tectonic processes, and
- *preservation bias*, where buoyant, felsic, and thermally stable crust is more likely to survive.

In steady-state geotherms, these processes matter because crustal heat production controls the *curvature* of the geotherm:

$$T(z) = T_0 + \frac{q_0}{k}z - \frac{A}{2k}z^2.$$

Two regions with identical surface heat flow can therefore have very different Moho temperatures, lithospheric thicknesses, and long-term stability depending on how heat-producing elements were redistributed and preserved through time.